Weak Convergence — Metric Space & Functional Analysis Vocabulary

Many ideas in weak convergence build on the structure of a **metric space**. The key is to keep concepts such as compactness, separability, completeness, boundedness, continuity, and measurability distinct.

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1. Metric Space

A **metric space** is a set $\mathbb{D}$ together with a distance function

$$d : \mathbb{D} \times \mathbb{D} \to [0, \infty).$$

The metric tells us how far apart two elements are.

Basic Example

For $\mathbb{R}$,

$$d(x, y) = |x - y|.$$

Functional-Analysis Example

For functions $f, g : T \to \mathbb{R}$,

$$d(f, g) = \|f - g\|_\infty = \sup_{t \in T} |f(t) - g(t)|.$$

This is the natural metric on spaces such as

$$\ell^\infty(T).$$

Topological Concepts

2. Open Ball

The open ball centered at x with radius ϵ is

$$B(x, \epsilon) = \{y : d(x, y) < \epsilon\}.$$

For example, in $\mathbb{R}$,

$$B(2, 0.5) = (1.5, 2.5).$$

Open balls are the basic building blocks of topology.

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3. Open Set

A set $G \subseteq \mathbb{D}$ is **open** if every point in G has some small ball around it that remains entirely inside G:

$$x \in G \implies \exists \epsilon > 0 : B(x, \epsilon) \subseteq G.$$

Example

$$(0, 1)$$

is open in $\mathbb{R}$.

Even if x is very close to 0 or 1, we can choose a sufficiently small ball around x that remains inside the interval.

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4. Closed Set

A set F is **closed** if it contains all of its limit points.

Equivalently, in a metric space,

$$x_n \in F, \qquad x_n \to x \quad \Longrightarrow \quad x \in F.$$

Example

$$[0,1]$$

is closed.

For example,

$$x_n = \frac{1}{n} \to 0,$$

and

$$0 \in [0,1].$$

Nonexample

$$(0,1]$$

is not closed because

$$\frac{1}{n} \to 0,$$

but

$$0 \notin (0,1].$$

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5. Closure

The **closure** of A, denoted

$$\overline{A},$$

is A together with all of its limit points.

Example

$$A = (0,1)$$

has closure

$$\overline{A} = [0,1].$$

An especially important example is

$$\overline{\mathbb{Q}} = \mathbb{R}.$$

This leads directly to the idea of **density**.

Sequences and Approximation

6. Convergence

A sequence x_n converges to x if

$$d(x_n, x) \to 0.$$

We write

$$x_n \to x.$$

Example

$$x_n = \frac{1}{n}$$

converges to 0, because

$$d(1/n, 0) = 1/n \to 0.$$

Function-Space Example

Under the sup norm,

$$f_n \to f$$

means

$$\sup_{t \in T} |f_n(t) - f(t)| \to 0.$$

This is **uniform convergence**.

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7. Cauchy Sequence

A sequence is **Cauchy** if its elements eventually become arbitrarily close to **each other**:

$$\forall \epsilon > 0, \quad \exists N : m, n > N \implies d(x_m, x_n) < \epsilon.$$

The definition does **not** require us to know the limit.

Mental Model

- **Convergent:** "I know where you are going."
- **Cauchy:** "You are clearly settling down somewhere."

Every convergent sequence is Cauchy.

Whether every Cauchy sequence converges depends on **completeness**.

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8. Complete

A metric space is **complete** if every Cauchy sequence converges to an element that is still inside the space:

$$\text{Cauchy} \quad \implies \quad \text{convergent in } \mathbb{D}.$$

Example

$$\mathbb{R}$$

is complete.

Nonexample

$$\mathbb{Q}$$

is not complete.

We can construct rational numbers q_n such that

$$q_n \to \sqrt{2}.$$

The sequence is Cauchy in $\mathbb{Q}$, but

$$\sqrt{2} \notin \mathbb{Q}.$$

Functional-Analysis Connection

A **Banach space** is simply a complete normed vector space.

Size and Covering Concepts

9. Bounded

A set A is bounded if it fits inside some ball of finite radius:

$$\exists x,\ M < \infty : A \subseteq B(x, M).$$

Example

$$[0, 1]$$

is bounded.

But

$$\mathbb{R}$$

is not.

Important: Bounded does not generally imply compact.

This becomes especially important in infinite-dimensional function spaces.

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10. Cover

A collection of sets

$$\{G_\alpha\}_{\alpha \in A}$$

is a **cover** of K if

$$K \subseteq \bigcup_{\alpha \in A} G_\alpha.$$

If every G_α is open, this is an **open cover**.

Example

The sets

$$(-1, 0.6)$$

and

$$(0.4, 1.1)$$

form an open cover of

$$[0, 1].$$

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11. Compact

A set K is **compact** if every open cover has a **finite subcover**.

Suppose

$$K \subseteq \bigcup_{\alpha \in A} G_\alpha.$$

Compactness says that we can find finitely many sets

$$G_{\alpha_1}, \dots, G_{\alpha_m}$$

such that

$$K \subseteq G_{\alpha_1} \cup \cdots \cup G_{\alpha_m}.$$

In $\mathbb{R}^d$

The Heine-Borel theorem gives

$$\boxed{K \text{ compact } \iff K \text{ closed and bounded.}}$$

Therefore,

$$[0, 1]$$

is compact.

But

$$(0, 1)$$

is not compact because it is not closed, and

$$[0, \infty)$$

is not compact because it is not bounded.

Important Warning

In infinite-dimensional spaces,

$$\boxed{\text{closed + bounded} \not\Rightarrow \text{compact.}}$$

This is one reason compactness becomes more subtle in functional analysis.

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12. Sequential Compactness

In metric spaces, compactness is equivalent to:

Every sequence in K contains a convergent subsequence whose limit is also in K.

If

$$x_n \in K,$$

then there exists a subsequence

$$x_{n_k}$$

such that

$$x_{n_k} \to x \in K.$$

This is often the easiest interpretation of compactness for probability theory.

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13. Totally Bounded

A set A is **totally bounded** if, for every $\epsilon > 0$, it can be covered by finitely many balls of radius ϵ:

$$A \subseteq \bigcup_{j=1}^{N} B(x_j, \epsilon).$$

Ordinary boundedness says the set fits inside **one sufficiently large ball**.

Total boundedness says that at **every resolution** ϵ, only finitely many ϵ-balls are needed.

A key result for metric spaces is

$$\boxed{\text{compact} \iff \text{complete + totally bounded.}}$$

This becomes important when studying **covering numbers** in empirical-process theory.

Density and Separability

14. Dense

A subset $D \subseteq \mathbb{D}$ is **dense** if every point in $\mathbb{D}$ can be approximated arbitrarily closely by elements of D.

Equivalently,

$$\overline{D} = \mathbb{D}.$$

Canonical Example

$$\mathbb{Q}$$

is dense in

$$\mathbb{R}.$$

For every $x \in \mathbb{R}$ and every $\epsilon > 0$, there exists $q \in \mathbb{Q}$ such that

$$|x - q| < \epsilon.$$

> Dense does **not** mean that D contains most of the points.

$\mathbb{Q}$ is countable while $\mathbb{R}$ is uncountable, yet $\mathbb{Q}$ is dense in $\mathbb{R}$.

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15. Separable

A metric space is **separable** if it contains a **countable dense subset**.

That is, there exists

$$D = \{d_1, d_2, ...\}$$

such that

$$\overline{D} = \mathbb{D}.$$

Canonical Example

$$\mathbb{R}$$

is separable because

$$\mathbb{Q}$$

is both countable and dense in $\mathbb{R}$.

Mental Model

> **Separable:** The entire space can be approximated arbitrarily well using only countably many representative points.

Do not confuse separability with compactness:

- **Compact:** finite subcover.
- **Separable:** countable dense subset.

Functions

16. Continuous

A function

$$f : \mathbb{D} \to \mathbb{E}$$

is continuous at x if

$$x_n \to x \quad \Longrightarrow \quad f(x_n) \to f(x).$$

The equivalent ϵ-δ definition is

$$\forall \epsilon > 0, \quad \exists \delta > 0$$

such that

$$d_{\mathbb{D}}(x, y) < \delta \implies d_{\mathbb{E}}(f(x), f(y)) < \epsilon.$$

Example

$$f(x) = x^2.$$

If

$$x_n \to x,$$

then

$$x_n^2 \to x^2.$$

For weak convergence, the sequential interpretation is often especially useful:

nearby inputs produce nearby outputs.

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17. Uniformly Continuous

Ordinary continuity allows the required δ to depend on the point x.

Uniform continuity requires one δ that works everywhere:

$$\forall \epsilon > 0, \quad \exists \delta > 0$$

such that, for every x, y,

$$d(x, y) < \delta \implies d(f(x), f(y)) < \epsilon.$$

A major result is

$$\boxed{f \text{ continuous on compact } K \Longrightarrow f \text{ uniformly continuous on } K.}$$

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18. Lipschitz

A function is **Lipschitz** if there exists $L < \infty$ such that

$$d(f(x), f(y)) \leq L\, d(x, y).$$

Example

$$f(x) = 2x$$

is Lipschitz with $L = 2$.

We have the hierarchy

$$\boxed{\text{Lipschitz} \Rightarrow \text{uniformly continuous} \Rightarrow \text{continuous.}}$$

Measure-Theoretic Concepts

19. σ-Algebra

A **σ-algebra** $\mathcal{A}$ is a collection of subsets of Ω that are allowed to be assigned measures or probabilities.

It must contain Ω and be closed under:

1. complements,
2. countable unions,
3. therefore also countable intersections.

Mental Model

> The σ-algebra specifies which subsets are legally measurable.

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20. Borel σ-Algebra

For a metric/topological space $\mathbb{D}$, the **Borel σ-algebra**

$$\mathcal{B}(\mathbb{D})$$

is the smallest σ-algebra containing all of the open sets.

Thus

$$\boxed{\text{Topology} \longrightarrow \text{Borel sets} \longrightarrow \text{Probability.}}$$

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21. Measurable Function

Consider

$$X : (\Omega, \mathcal{A}) \to (\mathbb{D}, \mathcal{B}).$$

The function X is **measurable** if

$$X^{-1}(B) \in \mathcal{A}$$

for every

$$B \in \mathcal{B}.$$

In words:

> Take a measurable set in the output space. Look at all inputs that map into that set. That preimage must also be measurable.

For a real-valued random variable, this can be checked using sets such as

$$\{\omega : X(\omega) \leq x\}.$$

A **random variable is therefore a measurable function**.

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22. Continuous vs. Measurable

Continuity and measurability are different properties.

For standard metric spaces,

$$\boxed{\text{continuous} \Rightarrow \text{Borel measurable.}}$$

But the converse is false:

$$\text{measurable} \not\Rightarrow \text{continuous}.$$

Example

$$f(x) = \mathbf{1}\{x \geq 0\}$$

is measurable but discontinuous at 0.

Probability Connections

23. Probability Measure

A probability measure

$$P : \mathcal{A} \to [0, 1]$$

assigns probabilities to measurable sets and satisfies

$$P(\Omega) = 1$$

and countable additivity.

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24. Tightness

A family of probability measures

$$\{P_\alpha\}$$

is **tight** if, for every $\epsilon > 0$, there exists a compact set K such that

$$P_\alpha(K) \geq 1 - \epsilon$$

for all α.

Mental Model

Almost all of the probability can be trapped inside one compact set.

This is why compactness becomes so important in weak-convergence theory.

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25. Weak Convergence

Roughly,

$$X_n \rightsquigarrow X$$

if

$$E[f(X_n)] \to E[f(X)]$$

for every bounded continuous function

$$f : \mathbb{D} \to \mathbb{R}.$$

One statement therefore combines much of the vocabulary above:

- metric spaces,
- boundedness,
- continuity,
- Borel measurability,
- probability measures,

- expectation.

When

$$\mathbb{D} = \ell^\infty(T),$$

additional issues such as

- compactness,
- separability,
- measurability,
- tightness,

become central.

Dependency Map

A useful way to organize these concepts is

$$\boxed{\text{Metric}}$$

$$\downarrow$$

$$\text{balls} \to \text{open sets} \to \text{closed sets / closure.}$$

From there,

$$\begin{array}{cc} \text{convergence} & \text{covers} \\ \downarrow & \downarrow \\ \text{Cauchy} & \text{compactness} \\ \downarrow & \downarrow \\ \text{completeness} & \text{total boundedness} \end{array}$$

Separately,

$$\text{dense} \to \text{separable.}$$

Topology connects to probability through

$$\text{open sets} \to \mathcal{B}(\mathbb{D}) \to \text{measurability} \to \text{probability measures.}$$

Finally,

$$\boxed{\text{compactness} + \text{probability} \to \text{tightness} \to \text{weak convergence.}}$$

Core Concepts to Memorize First

Concept	Trigger Phrase	Canonical Example
Closed	Contains its limits	$[0,1]$
Compact	Every open cover has a finite subcover	$[0,1]$
Dense	Can approximate every point	$\mathbb{Q} \subset \mathbb{R}$
Separable	Has a countable dense subset	$\mathbb{R}$, via $\mathbb{Q}$
Continuous	$x_n \to x \Rightarrow f(x_n) \to f(x)$	$f(x) = x^2$
Measurable	Preimages of measurable sets are measurable	Random variable
Complete	Every Cauchy sequence converges in the space	$\mathbb{R}$
Tight	Probability can be trapped in a compact set	$P(K) > 1 - \epsilon$

High-Value Relationships

$$\boxed{\text{compact} \iff \text{complete + totally bounded}}$$

for metric spaces.

$$\boxed{\text{compact} \iff \text{closed + bounded}}$$

in $\mathbb{R}^d$, but **not generally in infinite-dimensional spaces**.

$$\boxed{\text{Lipschitz} \Rightarrow \text{uniformly continuous} \Rightarrow \text{continuous} \Rightarrow \text{Borel measurable.}}$$

And perhaps the most important distinction to keep automatic:

> **Compactness = finite covering property. Separability = countable approximation property. Completeness = Cauchy sequences have limits. Measurability = inverse images are measurable.**